

PRAKTIKUM SISTEM OPERASI

MODUL 9
(SYSCALL)



Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

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TELKOM

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1. Buat syscall baru seperti yang ditunjukkan pada modul syscall poin 9.5!

```
#include <xinu.h>

syscall chname(pid32 pid, pri16 priority) {
    intmask mask;

    mask = disable();
    kprintf("New Syscall is easy\n");
    restore(mask);
    return OK;
}
```



The screenshot shows a terminal window titled 'C Tab Width: 8 Ln 16, Col 2 INS'. The terminal output is as follows:

```
Welcome to Xinu!

xsh $ uptime
New Syscall is easy
Uptime: 0 menit
xsh $
```

The status bar at the bottom of the terminal window reads: 'CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:0 | ttyS0'.

2. Perbaiki syscall chprio (xinu/system/chprio.c) dengan memperhatikan validasi input

```

chprio.c
~/xinu/system

*/
syscall chprio(
    pid32      pid,      /* ID of process to change */
    pri16      newprio   /* New priority */
)
{
    intmask mask;        /* Saved interrupt mask */
    struct proctab *prptr; /* Ptr to process's table entry */
    pri16      oldprio;   /* Priority to return */

    mask = disable();
    if (pid < 0 || pid >= NPROC) {
        restore(mask);
        return SYSERR;
    }

    if (newprio <= 0) {
        restore(mask);
        return SYSERR;
    }

    prptr = &proctab[pid];
    oldprio = prptr->prprio;
    prptr->prprio = newprio;
    restore(mask);
    return oldprio;
}

```

```

Welcome to Xinu!

xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr  Stack Size Msg Content
-----
0 prnull      ready  0    0 0x005FDFFC 0x00FFFE4 8192 1 3d
1 ipout       wait  500   0 0x005FBFFC 0x005FBF30 8192 0 0d
2 netin       wait  500   0 0x005F9FFC 0x005F9ED0 8192 0 0d
4 Main process recv  20    3 0x005E7FFC 0x005E7F34 65536 0 0d
5 shell       recv  50    4 0x005D7FFC 0x005D79AC 8192 0 0d
6 ps         curr  20    5 0x005F7FFC 0x005F7FC4 8192 0 0d

xsh $ uptime
Uptime: 0 menit
xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr  Stack Size Msg Content
-----
0 prnull      ready  0    0 0x005FDFFC 0x00FFFE4 8192 1 3d
1 ipout       wait  500   0 0x005FBFFC 0x005FBF30 8192 0 0d
2 netin       wait  500   0 0x005F9FFC 0x005F9ED0 8192 0 0d
4 Main process recv  20    3 0x005E7FFC 0x005E7F34 65536 0 0d
5 shell       recv  33    4 0x005D7FFC 0x005D79AC 8192 0 7d
8 ps         curr  20    5 0x005F7FFC 0x005F7FC4 8192 0 0d

xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:1 | ttyS0

```

3. Testing perubahan prioritas proses

```

/* xsh_uptime.c - xsh_uptime */

#include <xinu.h>
#include <stdio.h>
#include <string.h>

/*-----
 * xsh_uptime - shell to print the time the system has been up
 *-----
 */
shellcmd xsh_uptime(int nargs, char *args[])
{
    chprio(5,3);
    uint32  days, hrs, mins, secs; /* Days, hours, minutes, and */
                                     /* seconds since system boot */
    uint32  secperday = 86400;      /* Seconds in a day */
    uint32  secperhr  = 3600;       /* Seconds in an hour */
    uint32  secpermin  = 60;        /* Seconds in a minute */

    /* For argument '--help', emit help about the 'uptime' command */

    if (nargs == 2 && strcmp(args[1], "--help", 7) == 0) {
        printf("Use: %s\n\n", args[0]);
        printf("Description:\n");
        printf("\tDisplays time since the system booted\n");
        printf("Options:\n");
        printf("\t--help\t display this help and exit\n");
        return 0;
    }
}

```

```

Welcome to Xinu!

xsh $ ps
Pid Name          State Prio Ppid Stack Base Stack Ptr  Stack Size Msg Content
-----
0 prnull          ready  0    0 0x005FD0FFC 0x00FFFE4 8192 1 3d
1 ipout           wait   500  0 0x005FBFFC 0x005FBF30 8192 0 0d
2 netin          wait   500  0 0x005F9FFC 0x005F9ED0 8192 0 0d
4 Main process    recv   20   3 0x005E7FFC 0x005E7F34 65536 0 0d
5 shell          recv   50   4 0x005D7FFC 0x005D79AC 8192 0 0d
6 ps            curr   20   5 0x005F7FFC 0x005F7FC4 8192 0 0d

xsh $ uptime
Uptime: 0 menit

xsh $ ps
Pid Name          State Prio Ppid Stack Base Stack Ptr  Stack Size Msg Content
-----
0 prnull          ready  0    0 0x005FD0FFC 0x00FFFE4 8192 1 3d
1 ipout           wait   500  0 0x005FBFFC 0x005FBF30 8192 0 0d
2 netin          wait   500  0 0x005F9FFC 0x005F9ED0 8192 0 0d
4 Main process    recv   20   3 0x005E7FFC 0x005E7F34 65536 0 0d
5 shell          recv   33   4 0x005D7FFC 0x005D79AC 8192 0 7d
8 ps            curr   20   5 0x005F7FFC 0x005F7FC4 8192 0 0d

xsh $

```

4. Testing chprio syscall yang telah diubah
 - a. (5, -3)

```

/* xsh_uptime.c - xsh_uptime */

#include <xinu.h>
#include <stdio.h>
#include <string.h>

/*-----
 * xsh_uptime - shell to print the time the system has been up
 *-----
 */
shellcmd xsh_uptime(int nargs, char *args[])
{
    chprio(5,-3);
    uint32 days, hrs, mins, secs; /* Days, hours, minutes, and */
                                /* seconds since system boot */
    uint32 secperday = 86400; /* Seconds in a day */
    uint32 secperhr = 3600; /* Seconds in an hour */
    uint32 secpermin = 60; /* Seconds in a minute */

    /* For argument '--help', emit help about the 'uptime' command */

    if (nargs == 2 && strcmp(args[1], "--help", 7) == 0) {
        printf("Use: %s\n", args[0]);
        printf("Description:\n");
        printf("\tDisplays time since the system booted\n");
        printf("Options:\n");
        printf("\t--help\t display this help and exit\n");
        return 0;
    }
}

```

C Tab Width: 8 Ln 13, Col 22 INS

```

2 netin      wait  500  0 0x005F9FFC 0x005F9ED0 8192 0 0d
4 Main process  recv  20  3 0x005E7FFC 0x005E7F34 65536 0 0d
5 shell      recv  50  4 0x005D7FFC 0x005D79AC 8192 0 7d
8 ps         curr  20  5 0x005F7FFC 0x005F7FC4 8192 0 0d
xsh $ uptime
Uptime: 0 menit
xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr Stack Size Msg Content
-----
0 prnull      ready  0  0 0x005FDFFC 0x00FFFE4 8192 1 3d
1 ipout       wait  500  0 0x005FBFFC 0x005FBF30 8192 0 0d
2 netin      wait  500  0 0x005F9FFC 0x005F9ED0 8192 0 0d
4 Main process  recv  20  3 0x005E7FFC 0x005E7F34 65536 0 0d
5 shell      recv  50  4 0x005D7FFC 0x005D79AC 8192 0 9d
10 ps        curr  20  5 0x005F7FFC 0x005F7FC4 8192 0 0d
xsh $ uptime
Uptime: 0 menit
xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr Stack Size Msg Content
-----
0 prnull      ready  0  0 0x005FDFFC 0x00FFFE4 8192 1 3d
1 ipout       wait  500  0 0x005FBFFC 0x005FBF30 8192 0 0d
2 netin      wait  500  0 0x005F9FFC 0x005F9ED0 8192 0 0d
4 Main process  recv  20  3 0x005E7FFC 0x005E7F34 65536 0 0d
5 shell      recv  50  4 0x005D7FFC 0x005D79AC 8192 0 11d
12 ps        curr  20  5 0x005F7FFC 0x005F7FC4 8192 0 0d
xsh $

```

CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:39 | ttyS0

b. (3000,3)

```

/* xsh_uptime.c - xsh_uptime */

#include <xinu.h>
#include <stdio.h>
#include <string.h>

/*-----
 * xsh_uptime - shell to print the time the system has been up
 *-----
 */

shellcmd xsh_uptime(int nargs, char *args[])
{
    chprio(3000,3);
    uint32 days, hrs, mins, secs; /* Days, hours, minutes, and */
                                /* seconds since system boot */
    uint32 secperday = 86400; /* Seconds in a day */
    uint32 secperhr = 3600; /* Seconds in an hour */
    uint32 secpermin = 60; /* Seconds in a minute */

    /* For argument '--help', emit help about the 'uptime' command */

    if (nargs == 2 && strcmp(args[1], "--help", 7) == 0) {
        printf("Use: %s\n\n", args[0]);
        printf("Description:\n");
        printf("\tDisplays time since the system booted\n");
        printf("Options:\n");
        printf("\t--help\t display this help and exit\n");
        return 0;
    }
}

```

C Tab Width: 8 Ln 13, Col 22 INS

```

2 netin      wait  500  0 0x005F9FFC 0x005F9ED0 8192 0 0d
4 Main process  recv  20  3 0x005E7FFC 0x005E7F34 65536 0 0d
5 shell      recv  50  4 0x005D7FFC 0x005D79AC 8192 0 7d
8 ps         curr  20  5 0x005F7FFC 0x005F7FC4 8192 0 0d
xsh $ uptime
Uptime: 0 menit
xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr  Stack Size Msg Content
-----
0 prnull      ready  0  0 0x005FDFFC 0x00FFFE4 8192 1 3d
1 ipout       wait  500  0 0x005FBFFC 0x005FBF30 8192 0 0d
2 netin      wait  500  0 0x005F9FFC 0x005F9ED0 8192 0 0d
4 Main process  recv  20  3 0x005E7FFC 0x005E7F34 65536 0 0d
5 shell      recv  50  4 0x005D7FFC 0x005D79AC 8192 0 9d
10 ps         curr  20  5 0x005F7FFC 0x005F7FC4 8192 0 0d
xsh $ uptime
Uptime: 0 menit
xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr  Stack Size Msg Content
-----
0 prnull      ready  0  0 0x005FDFFC 0x00FFFE4 8192 1 3d
1 ipout       wait  500  0 0x005FBFFC 0x005FBF30 8192 0 0d
2 netin      wait  500  0 0x005F9FFC 0x005F9ED0 8192 0 0d
4 Main process  recv  20  3 0x005E7FFC 0x005E7F34 65536 0 0d
5 shell      recv  50  4 0x005D7FFC 0x005D79AC 8192 0 11d
12 ps         curr  20  5 0x005F7FFC 0x005F7FC4 8192 0 0d
xsh $

```

CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:39 | ttyS0